Warm-Up Answer Key

Ratios and Rates

LESSON 3-1 · UNDERSTAND RATIOS

MOVE *Add the parts to find the total / Write one part as a fraction of the total*
SLIDE 1) 24 2) 8 animals 3) 10, 12 4) 3/8
MODEL 8 3 3/8

1. 10 pens, 4/10	**5.** 10 cubes, 3/10
2. 8 pieces of fruit, 5/8	**6.** 3/8, 5/8
3. 12 vehicles, 5/12	**7.** 8 boys, 8/20
4. 15 players, 6/15	**8.** 5 cups, 3/5

DROPPED *skill 1 — bare multiplication-fact recall; it is rehearsed in box 1 of the warm-up slide and is not a step in the part-whole work this lesson assumes*

LESSON 3-2 · RATES AND UNIT RATES

MOVE *Divide to find the amount for one*
SLIDE 1) 15 2) 25, 30 3) 60 miles 4) $2
MODEL 4 15 15

1. $3	**5.** $1.50
2. 8 marbles	**6.** 7.5 liters
3. 50 miles	**7.** 9 boxes
4. 12 pages	**8.** $3 per pound, $2 per pound, Brand B

DROPPED *skill 3 — continuing a repeating pattern in a table is Lesson 3-3's move; a unit rate needs one division, not a table*

LESSON 3-3 · EQUIVALENT RATIOS USING TABLES

MOVE *Multiply both numbers of a ratio by the same factor*
SLIDE 1) 10 and 15 2) 16, 20 3) 6 4) 8 cups
MODEL 4 12 12

1. 15	**5.** 35
2. 15	**6.** 24
3. 8	**7.** 56, factor of 7
4. $16	**8.** 6, 9, 15

LESSON 3-4 · EQUIVALENT RATIOS USING GRAPHS

MOVE *Write and read an ordered pair by counting right, then up / Read what one gridline is worth before you plot*
SLIDE 1) (3, 2) 2) 5 3) 12 4) 24
MODEL 3 2 15

1. (5, 4)	**5.** 4 gridlines
2. (2, 7)	**6.** (20, 15)
3. 3	**7.** 3 right, 2 up
4. 12	**8.** (1, 30) and (2, 60)

DROPPED *skill 4 — generating the pairs with a ratio table is Lesson 3-3's move and is rehearsed there; in this lesson the pairs are already given*

LESSON 3-5 · COMPARING RATIO RELATIONSHIPS

MOVE *Scale both parts of a ratio to reach a target number, then compare*
SLIDE 1) 12 2) 4 3) 10 4) 10 pages
MODEL 4 16 16

1. 25
2. 21
3. 16
4. 6 of 10, Team B
5. 64
6. 8 cups, 9 cups, Recipe Y
7. 36, factor of 6
8. 21, 20, Juice B

DROPPED *skill 4 — dividing for a unit rate is Lesson 3-2's move and is rehearsed there; this worksheet stays on the common-term route so one move covers the page*

LESSON 3-6 · CONVERTING WITHIN ONE SYSTEM

MOVE *Multiply or divide by the conversion number*
SLIDE 1) 48 2) 12 inches 3) 3,000 4) 5
MODEL 48 100 250

1. 36 inches
2. 80 ounces
3. 2 feet
4. 2 gallons
5. 420 centimeters
6. 3.5 kilograms
7. 9 feet, 3 yards
8. 8 jars

LESSON 3-7 · CONVERTING BETWEEN SYSTEMS

MOVE *Multiply by the conversion factor and keep the unit label / Round the decimal answer to the nearest tenth*
SLIDE 1) 6.4 2) 4.5 3) 14 4) 11
MODEL 6.4 3.15 3.2

1. 8 kilometers
2. 1.8 kilograms
3. 7.62 centimeters
4. 13.2 pounds
5. 3.18 quarts, 3.2 quarts
6. 7.58 liters, 7.6 liters
7. 5 miles
8. 9 kilograms, 18 mL

DROPPED *skill 4 — chaining two conversions in one problem is what this lesson teaches; the warm-up only needs one factor applied cleanly, so it appears once as item 8, the bridge*